IST 691: Research Methodology and Scientific Writing Skills

Experimental Research Design

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Outline

- Introduction
- Faulty Research Design
- Requirements of True Research Designs
- Pretesting Participants
- True Research Designs

Introduction

- Research Design
  - The outline, plan, or strategy used to investigate a research problem
  - Purpose
    - To provide a plan in answering the research questions
  - Factors to consider
    - Type of Data to be Collected
    - Nature of the Data to be Analyzed
    - Research Questions to be Answered
    - Research Goal or Objective
  - Important consideration
    - Should maximize control over factors that could interfere with the validity of the findings

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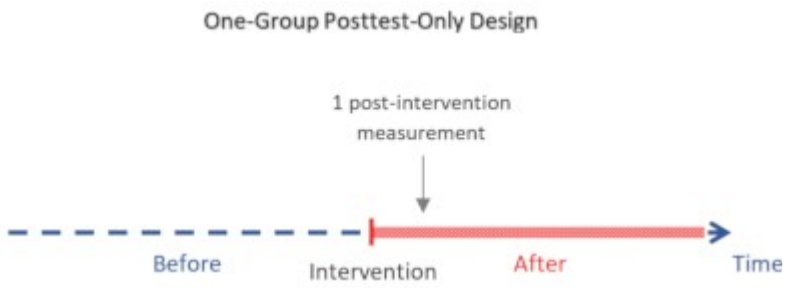
Faulty Research Design

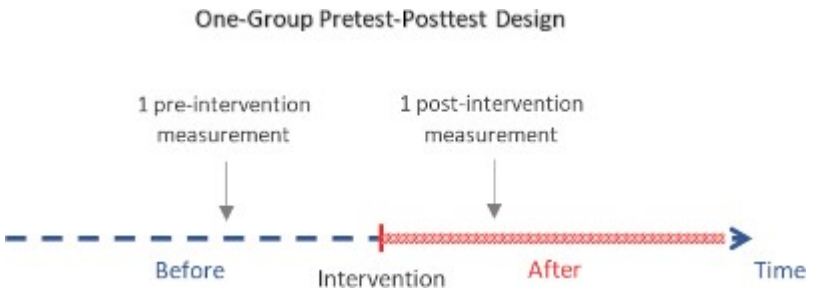
- One-Group After-Only Design
  - Also known as One-Group Post-test Only Design
  - A faulty research design in which the outcome of interest is measured only once after exposing a non-random group of participants to a certain intervention
  - Usually characterized by having:
    - No control group
    - No measurements before the intervention
  - Example
    - Investigating the effect of a training programme
    - On completion of programme, performance is measured



- Good performance suggests that the programme was successful, and the validity of the programme thought to have been established
- The design was however, useless
  - Without a basis for comparison, there is no way to determine if the performance was as a result of the training programme or not

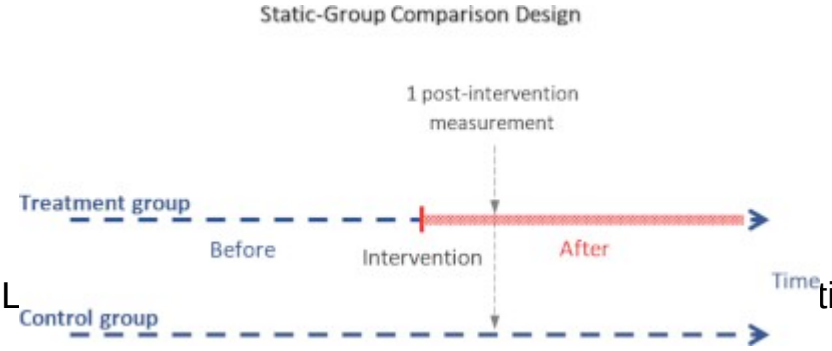
Faulty Research Design

- One-Group Before-After (Pretest-Post-test) Design
  - A faulty research design in which a treatment condition is interjected between a pre- and post-test of the dependent variable
  - Improvement over One-Group After-Only Design, but
    - Control group without the treatment condition for the purpose of comparison of treatment effect is missing
    - Especially important, as maturation may occur between the pretest and post-test



Non-equivalent Post-test-Only Design

- Primary disadvantage of the previous two designs
  - Impossibility of drawing any unambiguous conclusions as to the influence of the treatment condition
  - Non-equivalent post-test-only design attempts to remedy this deficiency by including a comparison group
    - One group of participants receive the treatment condition (X) and is compared on the independent variable (Y) with a group that did not receive the treatment condition



Non-equivalent Post-test-Only Design

- Design appears to be adequate
- The flaw, however, is that the performance of the experimental group is compared with that of a non-equivalent control group
- Non-equivalent in the sense that
  - The 2 groups are *assumed* to be equated on variables other than the independent variable
  - Even if matching is done on relevant variables such as age, gender, IQ, etc., matching is no assurance of having equated groups
    - The only way that one can have any assurance that the groups are equated is to assign the participants randomly to the experimental and the control groups
  - In studies where it is not possible to assign participants randomly, we are doomed – matching is the next best technique

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Requirements of True Research Designs

- Three criteria
  - Does the design answer the research question, or adequately test the hypothesis?
  - Have extraneous variables been controlled?
  - Generalizability (i.e., external validity)

Requirements of True Research Designs

- Does the design answer the research question, or adequately test the hypothesis?
  - Researcher must ask
    - What conclusion(s) can I draw from the experiment?
    - Note
      - The design of the study suggests the statistical tests that can be performed on the data, which in turn determine what conclusions can be drawn

Requirements of True Research Designs

- Have extraneous variables been controlled? (Internal validity)
  - To achieve internal validity, we must eliminate potential rival hypotheses
  - This can be achieved by using
    1) Control techniques
       - We saw control techniques in the previous chapter
       - The most important of these techniques is randomization – the only means by which unknown variables can be controlled
    2) A control group

Control Group and Experimental Group

- Control group
  - The group of research participants that serves as a standard of comparison for determining if the treatment condition produced any effect
  - Does not receive the independent variable (treatment condition) or receives a value that is considered a standard value
- Experimental group
  - Group of research participants that receives some amount of the independent variable

Control Group and Experimental Group

- Why a control group
  - Serves as a source of comparison
  - Serves as a control for rival hypotheses
    - All variables operating on the control and experimental groups must be identical, except for the the one being manipulated by the experimenter
    - In this way, the influence of the extraneous variables is held constant
- Note
  - Participants in the control group must be similar to participants in the experimental group
    - Theoretically, participants in both groups should yield identical scores in the absence of the introduction of the independent variable

Requirements of True Research Designs

- Generalizability (i.e., external validity)
  - Can the results of this experiment be applied to individuals other than those who participated in this experiment?
  - If yes, to whom do the results apply: only females, only females who attend college, etc.

Requirements of True Research Designs

- The three criteria we have examined:
  - Does the design answer the research question, or adequately test the hypothesis?
  - Have extraneous variables been controlled (internal validity)
  - Generalizability (i.e., external validity)
- These are the ideal
  - All studies must meet the first criterion
  - The degree to which the second and third are met varies between studies
    - Basic research usually focuses on internal validity
    - Applied research tends to focus on both internal validity and external validity since the central interest of such research is to apply the results to people and institutions

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Pretesting Participants

- Pre-testing
  - This is one means of obtaining information about the pre-treatment condition of the research participants
  - Experimenter can then directly observe change in participants' behaviour as a result of the treatment effect
  - But is pretesting essential in determining change in performance, when random assignment would already assign participants to control and experimental groups that can then be compared?

Pretesting Participants

- Several reasons to pretest

1) Increased sensitivity

   Can be achieved by matching participants on relevant variables

   Such matching requires pre-testing (see previous chapter)

2) Ceiling effect

   - By chance, all participants in an experiment may already hold (extreme) views that would not be affected by treatment
   - Pretesting will reveal this to the researcher

3) Initial position: some experiments require knowledge of the initial position (measure) of the subjects on the dependent variable, because it may interact with the experimental condition

4) Initial comparability

   Random assignment is normally used to ensure that participants are initially comparable on the relevant variables

   But random assignment, is not infallible and comparison of the subgroups' pretest mean scores would reveal this

5) Evidence of change: to gain an empirical demonstration of whether the treatment condition succeeded in producing the change in the research participants

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True Research Designs

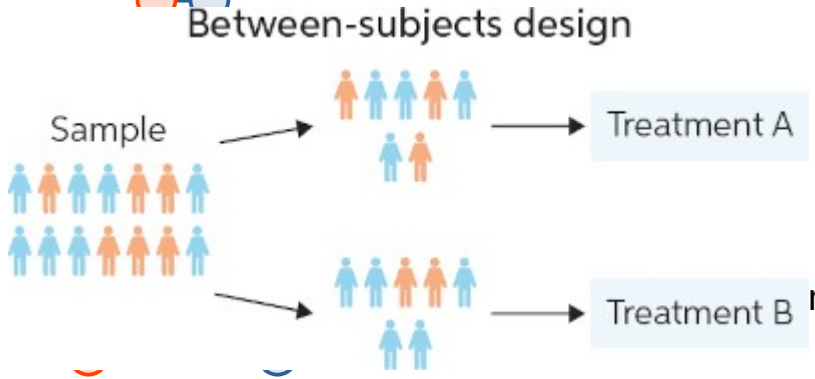
- True experimental design
  - A good experimental design in which the influence of extraneous variables is controlled for while the influence of the independent variable is tested
  - Types
    - After-only design
    - Before-after design

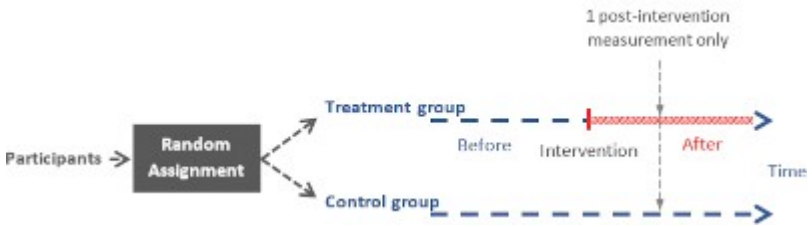
After-Only Design (Post-test Only Control Group Design)

- True experimental design in which the experimental and the control groups' post-test scores are compared to assess the influence of the treatment condition
- Exact structure of the design depends on several factors
  - Number of independent variables
  - Number of levels of variation of each independent variable
  - Whether the same or different participants are to be used in each treatment condition
    - If different participants are used in each experimental treatment condition, then the design is a *between-participants design*
    - If the same participants are used in each experimental treatment condition, then the design is a *within-participants design*

Between-Participants After-Only Design

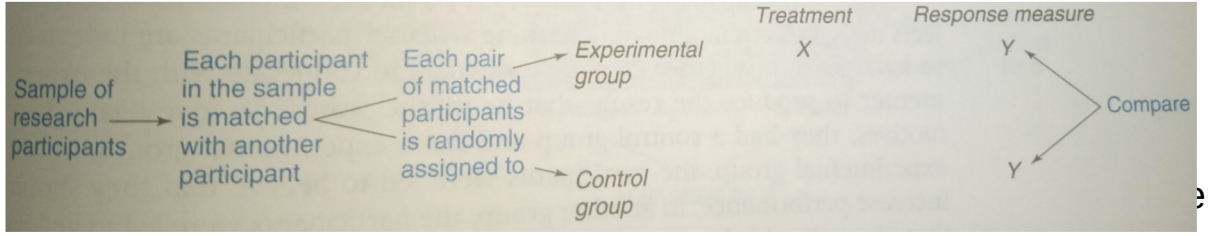
- After-only research design in which participants are randomly assigned to the experimental and control groups
- Unlike the non-equivalent post-test-only design, participants are assigned randomly to the control and treatment groups





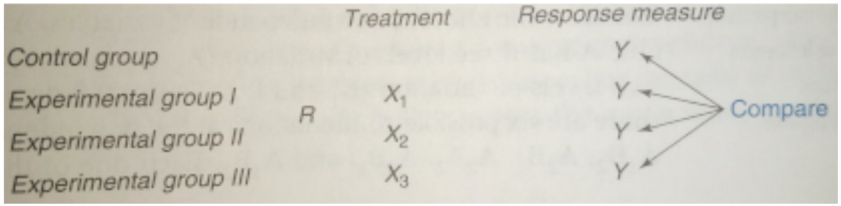
Matched Between-Participants After-Only Design

- Because randomization is not infallible, and to increase the sensitivity in the detected effect of the treatment, participants can be matched prior to random assignment
- Basic structure shown below is for two-group between-participants design
- But experiments are seldom confined to 2 levels of variation of one or more independent variables
- 2 primary extensions of the between-participants after-only design
  - Simple randomized participants design
  - Factorial design



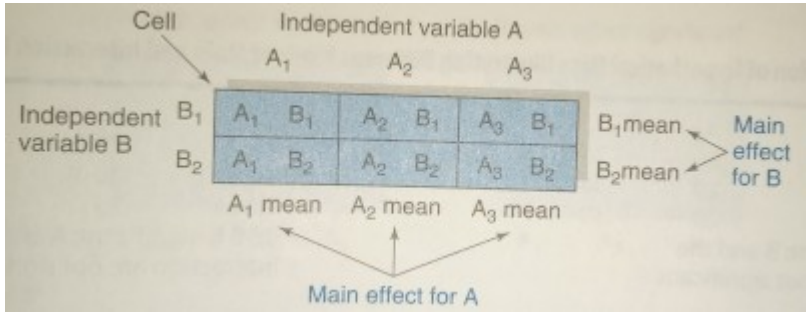
Simple Randomized Participants Design

- A between-participants design in which the influence of several levels of variation of the independent variable is investigated
- In drug research for example, investigator may want to administer different amounts of a drug to see if they produce differential reactions to the dependent variable



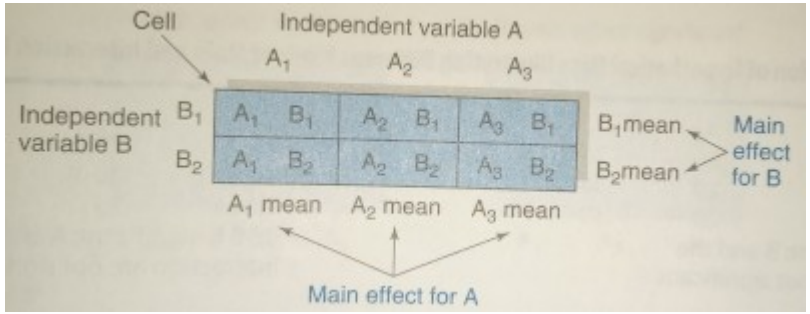
Factorial Design

- Simple randomized participants design considers only one independent variable
- In an experiment, we are usually interested in the effects of several independent variables acting in concert
- Factorial design
  - A between-participants design that enables us to investigate the independent and interactive influences of more than one independent variable



Factorial Design

- In the example
  - 2 independent variables: A and B
    - 3 levels for variable A (A_1, A_2, A_3)
    - 2 levels for variable B (B_1, B_2)
  - 6 possible combinations of the independent variables
    - A_1B_1, A_1B_2, A_1B_3, A_2B_1, A_2B_2, A_2B_3
  - Cell: A specific treatment condition in a factorial design – A_1B_1, A_1B_2, etc.

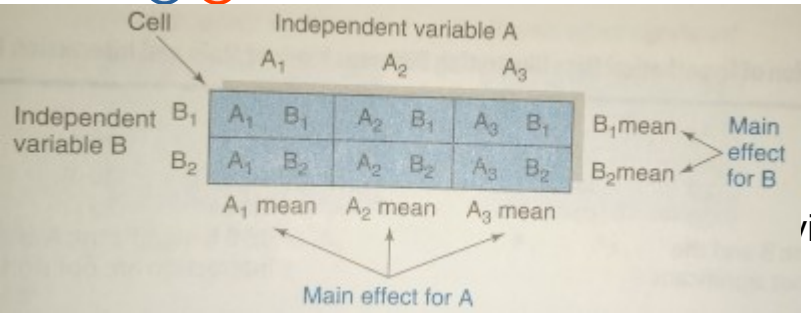


Factorial Design

- Two types of effects in factorial design: main effect and interaction effect
- Main effect
  - The overall effect of a single independent variable on the dependent variable, averaging across all levels of other factors
    - It indicates whether a specific factor significantly influences the outcome, ignoring other variables
    - It represents the direct, independent impact of that factor

Factorial Design

- Main effect
  - Was not discussed in simple randomized experiment or in the two-group between-participant after-only experiment because only one main effect of one independent variable existed
  - In a factorial design (which involves more than one independent variable) the separate effects of each independent variable must be identified



- In the example on the left, both A and B have main effects
- Main effect for A tells us
  - If A produced a significant influence on the dependent variable
  - Or, if there there was a significant difference among the three A mean scores
- Similarly, main effect for B tells us
  - If B produced a significant influence on the dependent variable
  - Or, if there there was a significant difference among the two B mean scores

Factorial Design

- Interaction effect
  - Influence of one independent variable on another independent variable
    - An interaction occurs when the effect of one factor on the dependent variable depends on the level of another factor, rather than being consistent across all conditions
    - It identifies whether factors work together, rather than just independently

Understanding Interaction

	A_1	A_2	A_3	Mean
B_1	10	20	30	20
B_2	10	20	30	20
Mean	10	20	30	

(*a*) A is significant; B and the interaction are not significant

	A_1	A_2	A_3	Mean
B_1	20	20	20	20
B_2	30	30	30	30
Mean	25	25	25	

(*b*) B is significant; A and the interaction are not significant

	A_1	A_2	A_3	Mean
B_1	30	40	50	40
B_2	50	40	30	40
Mean	40	40	40	

(*c*) Interaction is significant; A and B are not significant

	A_1	A_2	A_3	Mean
B_1	10	20	30	20
B_2	40	50	60	50
Mean	25	35	45	

(*d*) A and B are significant; interaction is not significant

	A_1	A_2	A_3	Mean
B_1	20	30	40	30
B_2	30	30	30	30
Mean	25	30	35	

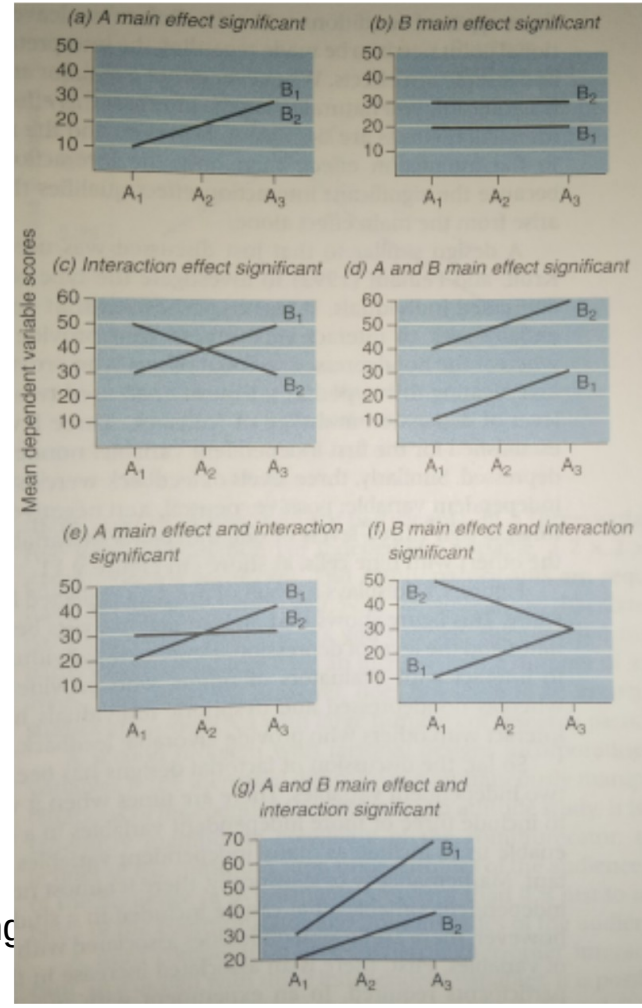
(*e*) A and the interaction are significant; B is not significant

	A_1	A_2	A_3	Mean
B_1	10	20	30	20
B_2	50	40	30	40
Mean	30	30	30	

(*f*) B and the interaction are significant; A is not significant

	A_1	A_2	A_3	Mean
B_1	30	50	70	50
B_2	20	30	40	30
Mean	25	40	55	

(*g*) A, B, and the interaction are significant



Writing

Understanding Interaction

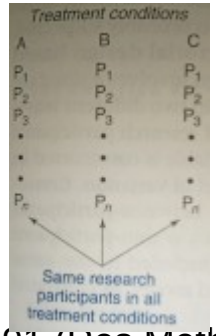
- (a), (b), (d)
  - One or more of the main effects are significant
    - Mean scores for the level of variation of at least one of the main effects differ
    - Notice lines for levels B_1 and B_2 are parallel in these examples
      - Interaction cannot exist because interaction means that the effect of one variable, e.g., B_1, depends on the level of the other variable being considered e.g., A_1, A_2, A_3
- (c)
  - Depicts classic example of interaction
  - Neither main effect is significant: the three column means are identical and the two row means are also identical, and reveal no variation
  - However, if effect of treatment A is considered only for level B_1, we notice that scores systematically increase form A_1 to A_3
  - Similarly, if only level B_2 is considered, then there is a systematic decrease from level A_1 to A_3
  - In other words, A is effective but in opposite directions for levels B_1 and B_2, i.e., the effect of A depends on which level of B we are considering

Understanding Interaction

- (e) and (f)
  - One of the main effects and interaction are significant
- (g)
  - Both main effects and interaction are significant
- Some notes
  - When either a main or an interaction effect ***alone*** is significant, you need to interpret this effect
  - When ***both*** interaction and main effects are significant, the main effect is contained in the interaction effect
    - Only the interaction effect is interpreted because the significant interaction effect qualifies the meaning that would arise from the main effect alone

Within-Participants After-Only Design

- A type of after-only design in which the same research participants are repeatedly assessed on the dependent variable after participating in all experimental treatment conditions
- This is a repeated measures design as all participants are repeatedly measured under each treatment condition



Advantages of Within-Participants After-Only Design

- Investigator need not worry about creating equivalence in research participants
  1) The same participants are involved in each treatment condition, and so serve as their own control, and variables such as age, gender, prior experience remain constant over the course of the experiment
  2) Does not require as many participants as between-participants design – same number as participants in a single group in between-participants design

Disadvantages of Within-Participants After-Only Design

- Confounding influence of a sequencing effect as the same participants participate in all the treatment conditions
  - If sequencing effects are not linear, then confounding influence of sequencing effects still exists, even with counterbalancing
- Because disadvantages far outweigh advantages, within-participants after-only design is not commonly used